MCS-215	MULTIMEDIA SYSTEMS	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To learn and understand technical aspect of Multimedia Systems.
CO2	To understand the standards available for different audio, video and text applications
CO3	To Design and develop various Multimedia Systems applicable in real time.
CO4	To learn various multimedia authoring systems
CO5	To understand various networking aspects used for multimedia applications.

Module 1

Introduction to Multimedia, Multimedia Objects, Multimedia in business and work. Multimedia hardware, Memory & Storage devices, Communication devices, Multimedia software s, presentation tools, tools for object generations, video, sound, image capturing, authoring tools, card and page based authoring tools.

Text, Sound MIDI, Digital Audio, audio file formats, MIDI under windows environment Audio & Video Capture.

Module 2

Huffman Coding, Shannon Fano Algorithm, Huffman Algorithms, Adaptive Coding, Arithmetic Coding Higher Order Modeling. Finite Context Modeling, Dictionary based Compression, Sliding Window Compression, LZ77, LZW compression, Compression, Compression ratio loss less & lossy compression.

Digital Audio concepts, Sampling Variables, Loss less compression of sound, loss compression & silence compression.

Module 3

Multiple monitors, bitmaps, Vector drawing, lossy graphic compression, image file formatic animations Images standards, JPEG Compression, Zig Zag Coding.

Video representation, Colors, Video Compression, MPEG standards, MHEG Standard recent development in Multimedia.

Books & References :

1. Tay Vaughan Multimedia, Making IT Work Osborne McGraw Hill.

2. Buford Multimedia Systems Addison Wesley.
3. Agrawal & Tiwari Multimedia Systems Excel.
4. Mark Nelson Data Compression Book BPB.
5. David Hillman Multimedia technology and Applications Galgotia Publications.
6. Rosch Multimedia Bible Sams Publishing.
7. Sleinreitz Multimedia System Addison Wesley.
8. **James E Skuman Multimedia in Action Vikas.**

Course Outcome:

CO1	Developed understanding of technical aspect of Multimedia Systems.
CO2	Understand various file formats for audio, video and text media.
CO3	Develop various Multimedia Systems applicable in real time.
CO4	Design interactive multimedia software.
CO5	Apply various networking protocols for multimedia applications.